Theme : Automotive Technology & Robotics and Automation

TITLE : Smart Accident Detection & Emergency Response System

A smart safety system designed to get help to accident victims faster.

SADES
SMART ACCIDENT DETECTION
& EMERGENCY RESPONSE

Presented by : Team Alert

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Domain : Disaster Resilience, Public Health & Community Well-being

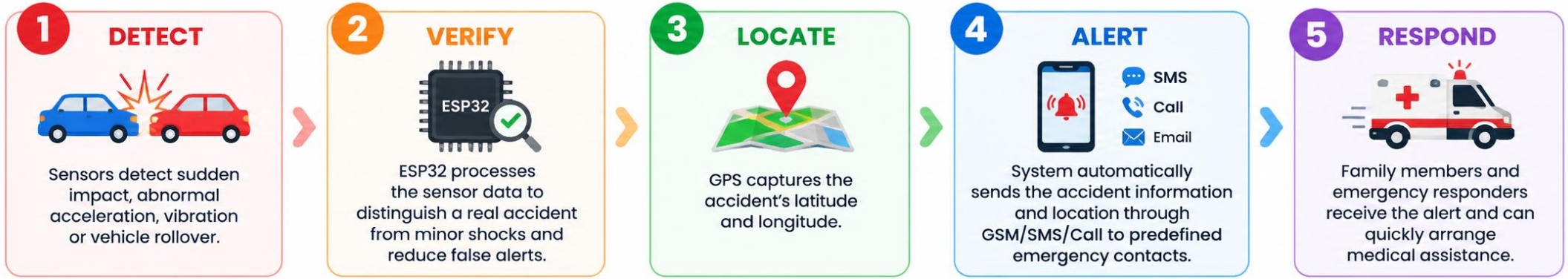
Problem statement

What happens after an accident? Accident occurs Victim may be unconscious Unable to make a call Accident is reported late. Location may be unknown Emergency response is delayed Problem: There is often a delay between accident occurrence and emergency notification.

Solution

SADES – Smart Accident Detection & Emergency Response System

A smart, low-cost system that automatically detects a vehicle accident, verifios the event, captures the exact location, and sends an emergency alert without depending on the victim to make a phone call.



CORE BENEFIT: "Detect the accident → Find the location → Send the alert → Save critical time."

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SMART ACCIDENT DETECTION & EMERGENCY RESPONSE SYSTEM

SADES – Smart Accident Detector & Emergency Response System

AUTOMATED ACCIDENT-RESPONSE MECHANISM

1. DETECT



Sensors continuously monitor vehicle movement and surroundings.

- Accelerometer
- Gyroscope
- Impact Sensor
- Sudden Deceleration Detection

ACCIDENT DETECTED

2. VERIFY



System analyzes sensor data to confirm an accident and reduce false alerts.

- Data Validation
- Threshold Check
- AI/Logic Confirmation

ACCIDENT VERIFIED

3. LOCATE



GPS module captures exact location of the accident.

- GPS Coordinates
- Time Stamp
- Location Accuracy

LOCATION ACQUIRED

4. ALERT



System sends alert with location to emergency contacts via GSM.

- SMS with Location Link
- Voice Call (Optional)
- Email Notification

ALERT SENT

5. RESPOND



Emergency services receive the alert and take immediate action.

- Nearest Help Dispatched
- Live Location Shared
- Faster Response

HELP ON THE WAY



All data securely stored in cloud for records and improvement.

CORE IDEA: Automatically detect an accident and quickly communicate the location to an emergency contact via GSM.

DETECT → VERIFY → LOCATE → ALERT → RESPOND

TECHNICAL APPROACH

SADES uses a combination of sensors, GPS, mobile communication, and cloud technology to detect accidents and send emergency alerts automatically.

Technical Workflow

1. **Accident Detection**
2. **Accident Verification**
3. **Location Detection**
4. **Emergency Alert**
5. **Emergency Response**

SADES
SMART ACCIDENT DETECTION & EMERGENCY RESPONSE SYSTEM

TECHNOLOGY STACK

HARDWARE	SENSORS	LOCATION	BACKEND	FRONTEND / ALERT	Communication
Microcoltroller (ESP32)	Accelerometer Gyroscope	GPS Module (Neo-6m)	Firebase (cloude database)	Mobile app / SMS / Call / Email	GMS / 5G (SIM800L)

SYSTEM WORKING FLOW

Accident Occurs → Sensors Detect Impact / Motion → ESP32 Processes Data → Location Fetched → Alert Sent via GSM → Contacts & Services Notified → Help & Response Arrives

VEHICLE IMPACT ENCOUNTERED

(Triggered by Accelerometer / Vibration Sensors)

REAL-TIME DATA PROCESSING EDGE

- Cross-checks G-force threshold limits
- Verifies vital physiological data (SpO2/Heart)
- Filters out minor bumps to prevent false alarms

AUTOMATED CLOUD & LOCATION DISPATCH

- Instantly fetches GPS coordinates (Lat / Long)
- Transmits incident data packet via IoT/GSM Cloud

SOCIETAL IMPACT

- Eradicates dependency on bystanders to report
- Lowers urban & highway mortality metrics
- Secure, continuous data logs for legal proof

EMERGENCY BENEFIT

- Triggers instant local hospital routing
- Transmits exact map coordinates directly
- Speeds up dispatch by bypassing manual calls

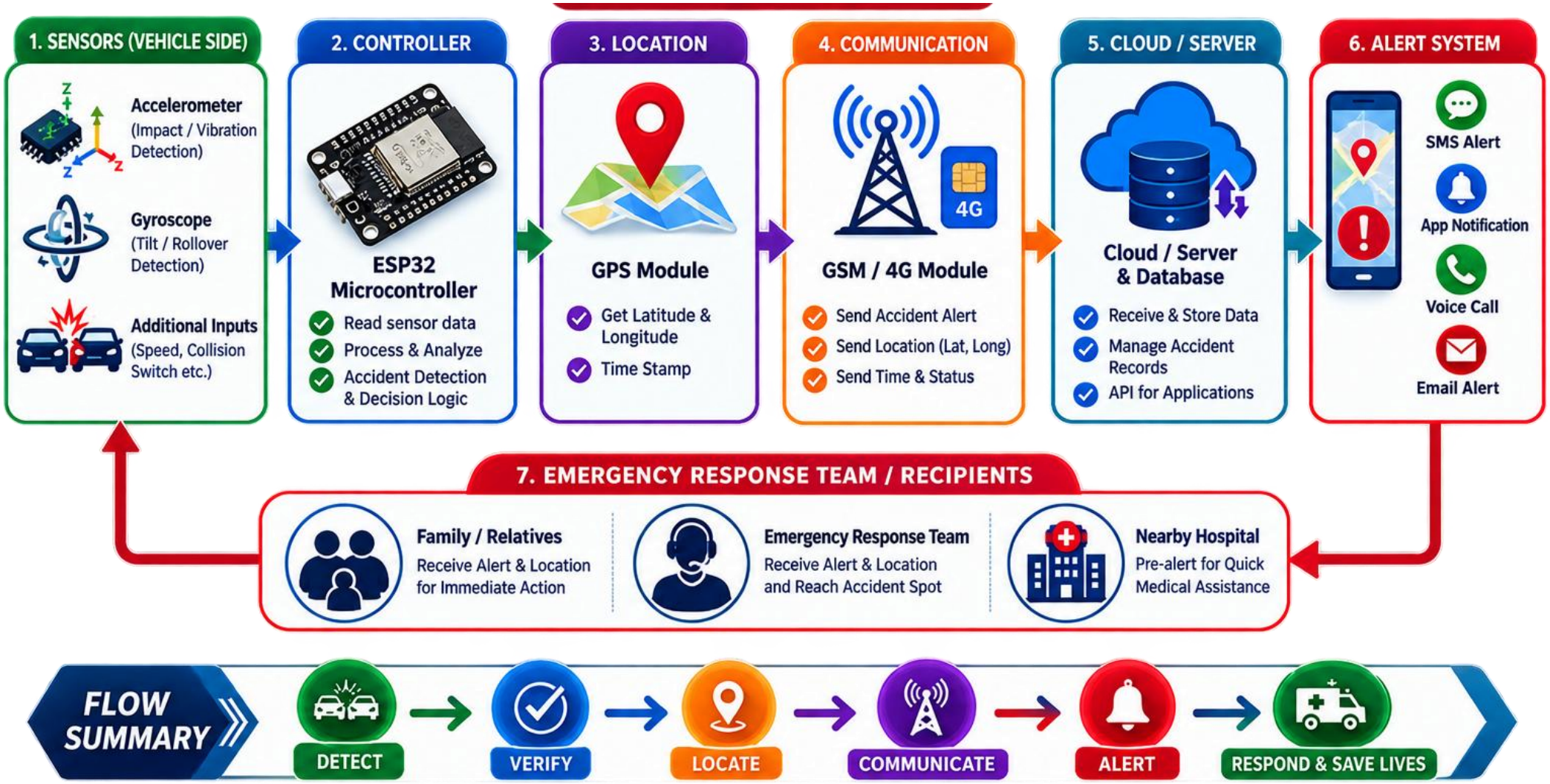
THE GOLDEN HOUR RESCUE WINDOW

- Rapid medical intervention preserves human life
- Optimized transit routes mitigate traffic blockages

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SMART ACCIDENT DETECTION & EMERGENCY RESPONSE SYSTEM

SADES
SMART ACCIDENT DETECTION & EMERGENCY RESPONSE SYSTEM

System architecture



FEASIBILITY AND VIABILITY

FEASIBILITY OF SADES:

1. Accident Detection
2. Emergency Alert
3. Fast & Low-Cost

CHALLENGES

1. Delayed Accident Detection
2. Accurate Accident Identification
3. Location & Network Issues
4. Fast Emergency Response

VIABILITY

1. Practical & Useful
2. Smart Technology
3. Low-Cost & Scalable+
4. Life-Saving Potential

FUTURE SCOPE

1. AI Integration
2. Emergency Connectivity
3. Mobile & Cloud
4. Smart Infrastructure

RESEARCH AND REFERANCES

Research:

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